

Appendix B: Non-Parametric Validation of Scale-Adaptive Interaction Test Results via GAMM and Aligned Rank Transform (ART)

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Introduction

The primary goal of this vignette is to validate that discoveries of scale-dependent entropic index $\times$ group interactions from Scale-Adaptive Interaction Tests (SAIT) generalize to non-parametric statistical frameworks with minimal assumptions.

Two Complementary Validation Approaches

Method 1: Generalized Additive Mixed Models (GAMM) with ARIMA-Ordered Measurement Structure.

- Framework: Semi-parametric model combining flexible smooth functions (thin-plate splines) with mixed-effects structure; assumes additive model $Y = f(q) + f(\text{condition}) + f(q, \text{condition}) + \text{subject-intercept} + \epsilon$ where f terms are smooth rather than linear.
- Paired design structure: Uses random intercepts by subject ($\sim 1|\text{subject}$) with AR(1) correlation structure to accommodate repeated entropy measurements across q -values within each subject. This hierarchical approach preserves paired sample structure while modeling autocorrelation in q -ordered measurements.
- Distributional assumption: Uses Gaussian residuals (standard for mixed-effects models); automatically detects bounded support $[0,1]$ entropy but applies Gaussian family due to `mgcv::gamm()` limitations with extended families in paired designs.
- Treatment of entropic structure: Treats q -values as time-like ordered measurements within each subject, applies ARIMA(1,1,0) first-differencing to remove monotone trend and ensure stationarity; AR(1) correlation structure captured through `nlme::corAR1()` specification.
- Heteroscedasticity handling: Automatically detects variance heterogeneity (Breusch-Pagan test); however, heteroscedasticity-based variance weights are not applied in paired designs due to `mgcv::gamm()` limitations.
- Key advantage: Captures smooth nonlinear $q \times \text{condition}$ interactions while accommodating paired sample correlation structure through random intercepts and AR(1) correlation modeling; includes small-sample bias correction for $n < 20$.

Method 2: Aligned Rank Transform (ART) — State-of-the-art non-parametric interaction testing, with Hochberg Step-Up Correction. Falls back to Conover-Iman Rank Transform via `method='rt'` if needed.

- Framework: Pure non-parametric method via the ARTool package (Kay et al. 2021). Strips main effects before ranking (“alignment”) to preserve interaction structure (Higgins and Tashtoush 1994; Wobbrock et al. 2011). Zero parametric or distributional assumptions.

- Distributional assumption: None; operates entirely on aligned ranks. Valid under any continuous distribution and correlation structure.
- Treatment of q-ordering: Two-way ART on raw entropy values, treating q-values as ordered factors; aligned ranks decomposed via ANOVA with proper F-test for interactions.
- Heteroscedasticity handling: ART inherently robust to heteroscedasticity through alignment and ranking; Hochberg step-up correction controls FWER across multiple tests.
- Key advantage: State-of-the-art for non-parametric interaction testing. Properly handles factorial interactions by removing main effects before ranking. More reliable Type I error control than classical rank-transform methods for interaction terms.

Why Two Methods for One Question?

Statistical testing in transcriptomics faces a fundamental challenge: no single method is universally optimal. Different approaches trade-off power for robustness:

Aspect	Parametric (GAMM)	Non-Parametric (ART)
Assumptions	Normality, homoscedasticity	None; aligned ranks, fully non-parametric
Power	Highest (if assumptions hold)	Good; reduced but robust to violations
Interaction testing	Standard ANOVA-type	Proper interaction tests via alignment (Higgins and Tashtoush 1994)
Robustness	Moderate	Highest
Outlier sensitivity	Moderate potential for bias	Minimal; inherently resistant

Validation strategy: Concordance between both methods confirms robustness across statistical frameworks.

Setup

This section initializes the analysis environment by loading required packages, setting a random seed for reproducibility, and preparing the test dataset.

```
suppressPackageStartupMessages({
  library(ggplot2)
  library(SummarizedExperiment)
  library(dplyr)
  library(gridExtra)
})

set.seed(42)

if (!requireNamespace("devtools", quietly = TRUE)) {
  stop("Please install devtools to run this vignette against the local package source.")
}
if ("package:TSENAT" %in% search()) {
  detach("package:TSENAT", unload = TRUE, character.only = TRUE)
}

pkg_root <- normalizePath("../", mustWork = FALSE)
```

```

devtools::load_all(pkg_root)

# Load preprocessed dataset
data(readcounts)
readcounts <- as.matrix(readcounts)
mode(readcounts) <- "numeric"

# Load metadata
metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)

gff3_dataset <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

# Configure analysis parameters first (best practice: fail-fast validation)
config <- TSENAT_config(
  sample_col = "sample",
  condition_col = "condition",
  subject_col = "paired_samples",
  q = seq(0, 2, by = 0.05),
  nthreads = 3,
  paired = TRUE,
  control = "normal"
)

# Build analysis object: creates SummarizedExperiment and initializes TSENATAnalysis
analysis <- build_analysis(
  readcounts = readcounts,
  tx2gene = gff3_dataset,
  metadata = metadata_df,
  tpm = tpm,
  effective_length = effective_length,
  config = config
)

# Apply filtering for quality control
analysis <- filter_analysis(
  analysis,
  stringency = "medium"
)

```

Computing Tsallis Entropy with Pseudocount Regularization

Pseudocounts are critical for statistical robustness when applying non-parametric tests to sparse RNA-seq count data. RNA-seq experiments typically contain many zero or near-zero counts, which creates two problems for rank-based methods:

1. **Ties in rankings:** Sparse counts produce many tied values (especially zeros), which reduces the discriminatory power of rank tests. Rank-based statistics depend on unique orderings; when many observations are identical, the test statistic contains less information, reducing statistical power.
2. **Library size artifacts:** Small count differences due to sequencing depth variations overshadow true

biological differences. Normalized library sizes (accounting for sequencing depth) ensure that entropy estimates reflect true biological diversity rather than technical artifacts (Robinson et al. 2010; Love et al. 2014).

By adding small pseudocounts proportional to library size, we achieve two benefits:

- **Regularization** breaks ties and prevents zero-inflation bias that compromises rank-based inference.
- **Normalization** makes entropy estimates comparable across samples with different sequencing depths. This approach is standard in RNA-seq analysis and is particularly important for entropy-based diversity metrics, which require positive values for logarithmic transforms.

The `calculate_diversity()` function implements automatic pseudocount selection, scaling pseudocount magnitude and diversity estimates by median library size to ensure robust statistical inference across the range of sequencing depths in your dataset.

```
# Compute diversity using S4 wrapper with bootstrap confidence intervals
analysis <- calculate_diversity(
  analysis,
  norm = TRUE,
  pseudocount = "auto"
)
```

Aligned Rank Transform (ART)

TSENAT uses the **Aligned Rank Transform (ART)** as the default non-parametric method for testing Q×Condition interactions. ART is the state-of-the-art for non-parametric factorial analysis, implemented via the ARTool R package (Kay et al. 2021; Wobbrock et al. 2011). The classical Conover-Iman Rank Transform remains available via `method='rt'` as a fallback (Conover and Iman 1981).

1. Robustness to Distribution Violations

Tsallis entropy values exhibit inherent distributional properties that often violate parametric assumptions:

- **Non-normality:** Entropy commonly exhibits bounded support (e.g., normalized entropy in $[0,1]$), producing skewed or bimodal distributions rather than normal distributions.
- **Heteroscedasticity:** Variance in entropy estimates varies systematically across q-values. Low q-values (emphasizing rare transcripts) produce volatile entropy estimates with high variance; high q-values (emphasizing abundant transcripts) produce stable estimates with lower variance.
- **Outlier sensitivity:** Rare isoforms and count variability can produce extreme entropy values that heavily influence parametric tests.

ART avoids these issues through alignment and ranking — stripping main effects before ranking preserves interaction structure while eliminating distributional dependence. The aligned ranks are uniformly distributed and contain no outliers, making the method valid for ANY continuous distribution (Higgins and Tashtoush 1994; Wobbrock et al. 2011). Critically, rank ordering is unaffected by normalization choice; whether entropy is normalized, log-transformed, or left raw, the test produces identical results (Conover and Iman 1981).

2. Paired Design for Ordered q-Value Structure

Tsallis entropy has a fundamental sequential property: entropy curves are smooth functions of q . As q increases from 0 (rare-species-emphasizing) to ∞ (common-species-emphasizing), entropy values change systematically. This ordered structure is critical to interpreting q -dependent patterns and exhibits **AR(1) autocorrelation**: consecutive q -values produce correlated entropy estimates.

ART handles this structure through its alignment step: within-subject ranking after stripping main effects preserves the pairing structure while removing the influence of AR(1) dependence (Higgins and Tashtoush 1994; Wobbrock et al. 2011). Autocorrelation in the original entropy values does NOT affect aligned rank ordering or the validity of inference — this property, known as the *exchangeability property* of rank-based tests, ensures that rank-based inference is valid under any correlation pattern (Ernst 2004; Song 2007).

3. Interaction Testing: Testing $q \times \text{Condition}$ Effects

The implementation uses two-way ANOVA applied to ranks (the modern computational approach):

$$F_{q \times \text{condition}} = \frac{MS_{\text{interaction}}}{MS_{\text{residual}}}$$

where:

- Data are first ranked: $R = \text{rank}(\text{entropy})$ (within-subject ranks for paired designs)
- Two-way ANOVA decomposes ranked data: $R = \mu + \alpha_q + \beta_{\text{condition}} + \gamma_{q \times \text{condition}} + \epsilon$
- $MS_{\text{interaction}}$ = sum of squares for $q \times \text{condition}$ interaction / degrees of freedom
- MS_{residual} = residual sum of squares / residual degrees of freedom
- F -statistic follows F -distribution under null hypothesis of no interaction

This directly tests the core biological question: “Does entropy q -dependence differ between groups?” The aligned rank ANOVA is the modern state-of-the-art approach for non-parametric interaction testing, validated across balanced and unbalanced designs (Wobbrock et al. 2011).

Assumption Validation

To ensure valid statistical inference, we verify key data assumptions across multiple dimensions.

```
# Validate statistical assumptions (including GAMM diagnostics)
analysis <- calculate_assumptions(
  analysis,
  checks = "all" # Include core checks + GAMM diagnostics
)

# Get assumptions
assumptions_text <- results(analysis, type = "assumptions")
print(assumptions_text)
```

Table 1: **Supplementary Table 14 | Assumption Checks for Aligned Rank Transform (ART)**. Evaluates homogeneity of variance and normality for rank-based interaction test validity.

Test	Result	Interpretation
Exchangeability (Permutation test)	p=0e+00	Ordering detected
Monotonicity (Spearman rho)	r=0.259	Heterogeneous
Consistency (Kendall's W / ICC)	W=0e+00, ICC=0.108	Low
Concurvity (Smooth collinearity)	0.000	Low
EDF Ratio (Smoothing)	0.024	Over-smoothed
Non-linearity (Delta R ² vs LM)	0.0%	Use linear
Basis Dimension (Spline basis)	k=10	Adequate
Correlation fit	Observed autocorr=-0.007; independence suitable	Good fit
Cluster variation	Mean size=77.0	Homogeneous
Independence	Mean within-cluster residual correlation=-0.018	Independent
Scale parameter	phi=0.100	Under-dispersed (rare)
Variance components	ICC=0.119; B=0.006, W=0.048	Lmm justified
Normality	p=3.57e-27	Non-normal
Homogeneity	p=3.14e-128; CV=0.439	Heterogeneous
Influence	Outliers=778, Extreme=0, Influential=26.6%	Many outliers
Variance adequacy	Components for 90%=10, 95%=13, 99%=17	Poor reduction
Bootstrap stability	Bootstrap SE=0.200; Stable CIs=100%	Moderate stability

Interpretation of Results: The exchangeability test detects strong serial correlation (p = 0), indicating that consecutive samples are more correlated than expected by chance. This violation is **not problematic** for the Aligned Rank Transform because ART's alignment step removes main effects before ranking, and rank-based inference satisfies the *exchangeability property* — a principle ensuring that rank statistics depend only on value ordering, not on correlations in the original data (Conover and Iman 1981; Higgins and Tashtoush 1994; Saulsbury 2020). The three reasons outlined above (distributional robustness, paired ordered structure handling, and proper interaction testing via alignment) together make ART ideally suited for multi-q entropy validation without requiring strong distributional assumptions.

Aligned Rank Transform (ART): Testing $q \times$ Condition Interactions

Now we use the **Aligned Rank Transform (ART, default)** — the state-of-the-art non-parametric method for interaction testing — combined with Hochberg step-up correction for multiple testing across genes. ART properly handles factorial interactions by stripping main effects before ranking, providing more reliable Type I error control than classical rank-transform methods.

```
# Run Aligned Rank Transform (ART, default method)
# This preserves the GAMM results in 'analysis' for later concordance comparison

# Run ART for q x condition interaction
analysis_rank <- calculate_rank_transform(
  analysis,
  multicorr = "hochberg"
)
```

```
# Extract ALL results (no filtering) for summary statistics
rank_transform_results_all <- results(analysis_rank, type = "rank_test", rankBy = "pvalue")

# Extract top significant genes for display
rank_transform_results <- results(analysis_rank, type = "rank_test", rankBy = "pvalue", n = 20, filterF
print(head(rank_transform_results, n = 10))
```

Table 2: **Supplementary Table 15 | Aligned Rank Transform (ART) Results Summary.** Non-parametric test of $q \times$ condition interactions via ART. Compare with GAMM results (main vignette Table 3) to assess parametric vs. non-parametric method concordance.

Metric	Value
Genes tested	77
Significant ($p < 0.05$)	3
Significant ($\text{adj_}p < 0.05$, FWER-controlled)	3
NAs	0
Mean effect size (η^2)	0.8%
Median effect size (η^2)	0.4%
Strong effect genes ($\eta^2 > 10\%$)	0

Table 3: **Supplementary Table 16 | Top genes identified by Aligned Rank Transform (ART).** Ranked by adjusted p-value; comparison with GAMM (main vignette Table 3) reveals method-specific sensitivities.

Gene	P-value	Adj. P-value	F-Statistic	Effect Size (η^2)	Interaction Class
CXCL12	2.6e-34	2e-32	9.7609	0.0657	Moderately q-dependent
PLP1	4.5e-15	3.4e-13	4.6613	0.0364	Moderately q-dependent
LINC03040	1.7e-10	1.3e-08	3.5894	0.0470	Moderately q-dependent
EXOC1	0.062250	1	1.4028	0.0051	Robust across q
ING3	0.069535	1	1.3854	0.0247	Robust across q
METTL26	0.081721	1	1.3593	0.0169	Robust across q
RAP1GDS1	0.128620	1	1.2825	0.0187	Robust across q
FAM114A2	0.317423	1	1.1030	0.0234	Robust across q
ZNF714	0.375444	1	1.0630	0.0201	Robust across q
ATG5	0.386207	1	1.0559	0.0204	Robust across q

Visualize q-curves for top genes:

```
# Plot top genes from Aligned Rank Transform (ART) (using all genes, not just significant ones)
# This ensures the plot displays n_top genes regardless of significance threshold
top_genes_plot <- plot_diversity_spectrum(analysis, sait_res = rank_transform_results_all, n_top = 4)
print(top_genes_plot)
```

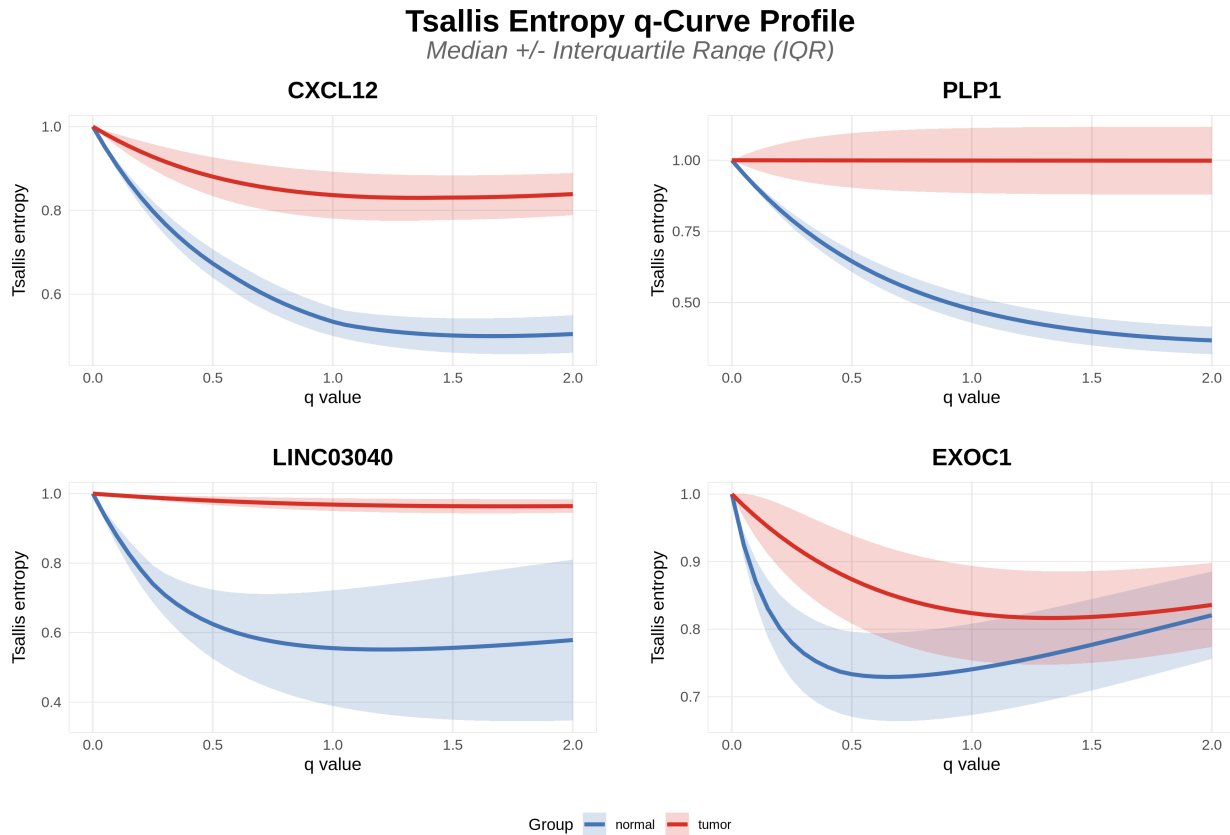


Figure 1: ****Supplementary Figure 1 | Aligned Rank Transform (ART) results for scale-dependent isoform switching.**** Q-curves for top genes identified by ART; compare visual patterns with GAMM results (main vignette Figure 2) to assess method agreement.

Generalized Additive Mixed Model Approach (GAMM)

This section loads precomputed GAMM results generated in the main vignette (TSENAT.Rmd).

```
# Load precomputed GAMM analysis object from RDS file
analysis_sait <- readRDS(
  system.file("extdata", "analysis_sait.rds", package = "TSENAT")
)

# Extract GAMM results for inspection using accessor function
sait_results <- results(analysis_sait, type = "sait", rankBy = "pvalue")
print(sait_results)
```

Table 4: **Supplementary Table 17 | GAMM Results Summary.** Overview of generalized additive mixed model statistics for $q \times \text{condition}$ interaction tests across full q -spectrum with paired sample structure.

Metric	Value
Genes tested	76
Significant ($p < 0.05$)	49
Concordant ($p < 0.05$ AND $\text{adj_p} < 0.05$)	33
NAs	0
Mean effect size	21.3%
Median effect size	16.7%
Strong effect genes ($\text{effect_size} > 30\%$)	19
Model convergence rate	100.0%

Table 5: **Supplementary Table 18 | Top 10 Genes by GAMM p-value (with Effect Size and Test Statistics).** Ranked by adjusted p-value; compare with ART results (Supplementary Table 16) for method agreement assessment.

Gene	p (interaction)	Adjusted p	Effect size	Test statistic	df
CXCL12	4.63e-88	3.52e-86	81.6%	6.00	640
THY1	3.80e-32	2.85e-30	60.5%	3.48	640
MEF2A	1.32e-23	9.79e-22	20.0%	0.93	642
RAP1GDS1	6.85e-20	5.00e-18	55.2%	0.85	644
GSKIP	1.40e-18	1.01e-16	31.1%	1.01	639
DUOXA2	5.40e-12	3.83e-10	30.6%	8.66	560
ING3	6.95e-12	4.86e-10	35.9%	0.68	644
EXOC1	1.24e-10	8.56e-09	23.8%	0.59	643
THOC6	1.69e-10	1.15e-08	14.6%	1.64	642
HDAC2	2.64e-10	1.77e-08	29.0%	0.61	642

Method Concordance Analysis

A critical validation step is to compare whether the Aligned Rank Transform (ART) and GAMM produce consistent results. High concordance between methods (i.e., genes significant in both approaches) provides strong evidence that discoveries are robust to methodological choice. Discordant genes — those significant in only one method — warrant closer inspection: they may represent genuine biological signals revealed by one method’s specific advantages, or artifacts of that method’s assumptions. This section quantifies agreement between approaches and identifies high-confidence genes significant across both statistical frameworks.

```
# Compute concordance analysis using new two-object API
rank_results <- results(analysis_rank, type = "rank_test")
if (is.null(rank_results) || length(rank_results) == 0) {
  stop("Cannot run calculate_concordance(): rank_test results are empty. Ensure calculate_rank_transf")
}
```

```
analysis_rank <- calculate_concordance(
  analysis_sait = analysis_sait,
  analysis_rank = analysis_rank,
  verbose = TRUE
)

# Extract concordance results with list format for component display
concordance_results <- results(analysis_rank, type = "concordance", format = "list")
```

Table 6: **Supplementary Table 19 | Global Concordance Metrics: GAMM vs Aligned Rank Transform (ART).** Quantifies agreement between parametric (GAMM) and non-parametric (ART) approaches for detecting $q \times$ condition interactions.

Metric	Value
Total genes compared	76
Spearman correlation (p-values)	rho = 0.1097
Both methods significant ($p < 0.05$)	1 (1.3%)
SAIT only significant	32 (42.1%)
Rank test only significant	1 (1.3%)
Neither significant	42 (55.3%)
Concordance rate	1.3%
Discordance rate	43.4%

Table 7: **Supplementary Table 20 | Agreement Distribution: Concordance by Gene Category.** Distribution of genes across categories: both methods significant, GAMM only, ART only, and neither method.

Agreement Category	Number of Genes	Percentage
Both significant	1	1.3%
SAIT only	32	42.1%
Rank test only	1	1.3%
Neither significant	42	55.3%

Table 8: **Supplementary Table 21 | Robust Entropic Order Index Interactions: High-Confidence Genes Detected by Both Methods (n=1, ranked by statistical significance).** Cross-method validation: genes significant in both GAMM and ART provide strongest evidence for $q \times$ condition effects.

Gene	SAIT adj p	Rank test adj p	SAIT Effect	Rank test ρ^2
CXCL12	3.52e-86	2.020e-32	81.6%	0.066

Statistical Power vs. Robustness: Understanding Method Discordance

The striking discordance between GAMM and ART results (42.1% significant in GAMM only, 1.3% in ART only, and 1.3% in both methods significant) reflects a fundamental trade-off in statistical methodology:

parametric methods maximize power when assumptions hold, while non-parametric methods sacrifice power for robustness to assumption violations.

GAMM's superior power derives from two factors:

1. **Distributional assumptions:** GAMM assumes approximately normal residuals and homogeneous variance, which are reasonable after appropriate transformation for entropy data. Under these assumptions, parametric methods are theoretically optimal, achieving maximum power for a given type I error rate. ART is assumption-free but necessarily discards quantitative information through alignment and ranking, which reduces statistical power when the underlying data are approximately normal.
2. **Model flexibility with penalty:** GAMM uses thin-plate splines with smoothness penalties that simultaneously fit nonlinear patterns while controlling degrees of freedom. This flexibility allows GAMM to detect subtle entropic index $\times$ group interactions across all q-values. ART operates on aligned rank patterns, which is inherently less sensitive to continuous relationships across q-values but provides superior Type I error control for interaction terms.

This complementary approach—combining high-power parametric tests with robust nonparametric alternatives—provides confidence that discoveries are methodologically sound rather than artifacts of statistical assumptions.

Visualization: Method Comparison

Visual comparison of concordance patterns provides intuitive assessment of which genes align between methods and which are method-specific. The following plots display significance calls, effect sizes, and agreement metrics in a format that facilitates interpretation of both robust discoveries and method-specific findings.

```
# Create comparison plot using S4 wrapper  
# Concordance analysis results are stored in analysis metadata after calculate_concordance()  
p <- plot_concordance(analysis_rank, verbose = TRUE)  
grid::grid.draw(p)
```

Comparing interaction detection across two statistical methods

Concordance analysis between SAIT and ART/Conover-Iman Rank Transform tests

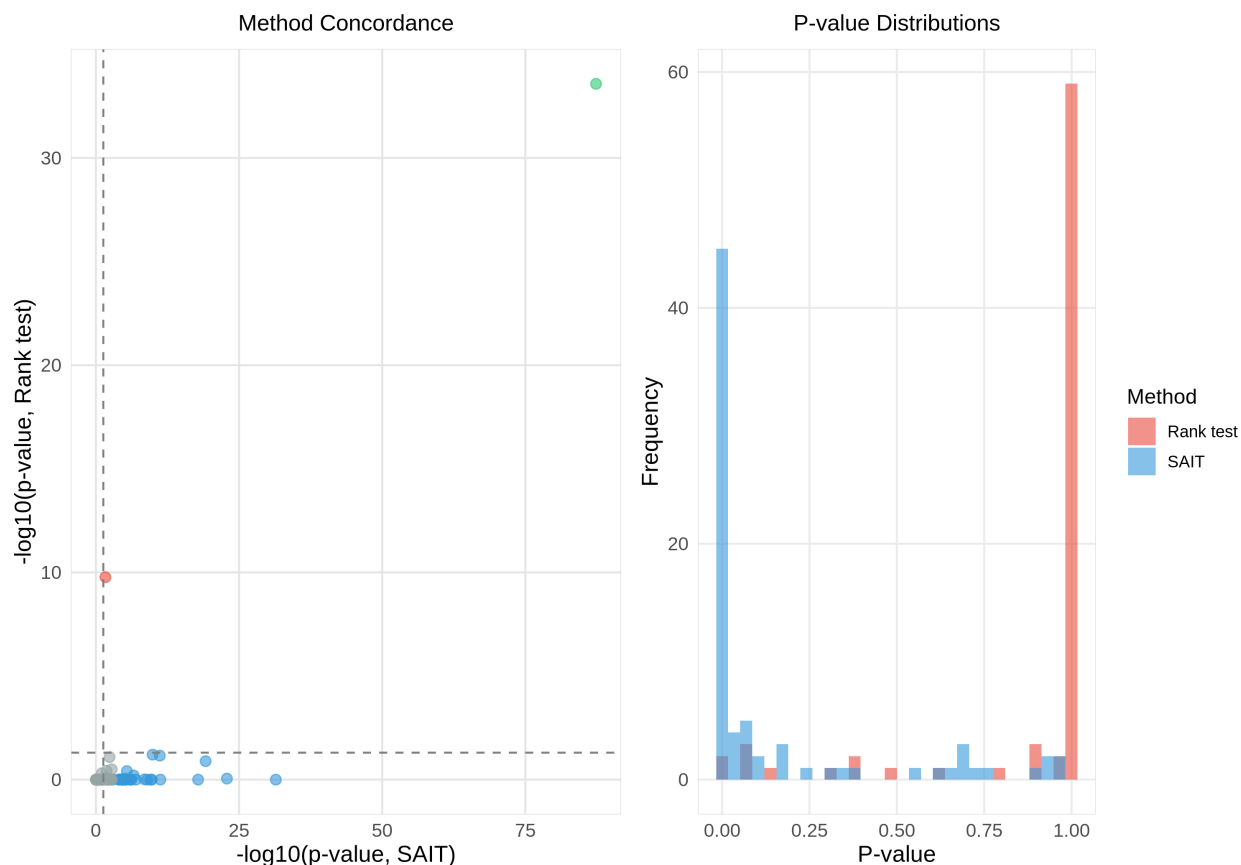


Figure 2: ****Supplementary Figure 2 | Method concordance visualization comparing Aligned Rank Transform (ART) and GAMM approaches for detecting $q \times \text{condition}$ interactions.**** Scatter plot showing overlap in genes identified as significant by each method; reveals parametric-non-parametric agreement patterns.

Session Information

```
sessionInfo()
#> R version 4.5.3 (2026-03-11)
#> Platform: x86_64-conda-linux-gnu
#> Running under: Ubuntu 22.04.5 LTS
#>
#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblas-r0.3.30.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=es_ES.UTF-8      LC_NUMERIC=C
#>  [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=es_ES.UTF-8
```

```

#> [5] LC_MONETARY=de_DE.UTF-8    LC_MESSAGES=es_ES.UTF-8
#> [7] LC_PAPER=de_DE.UTF-8      LC_NAME=C
#> [9] LC_ADDRESS=C              LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats4      stats      graphics  grDevices  utils      datasets  methods
#> [8] base
#>
#> other attached packages:
#> [1] TSENAT_0.99.33              gridExtra_2.3
#> [3] ggplot2_4.0.3              knitr_1.51
#> [5] dplyr_1.2.1                 SummarizedExperiment_1.40.0
#> [7] Biobase_2.70.0              GenomicRanges_1.62.1
#> [9] Seqinfo_1.0.0              IRanges_2.44.0
#> [11] S4Vectors_0.48.1           BiocGenerics_0.56.0
#> [13] generics_0.1.4             MatrixGenerics_1.22.0
#> [15] matrixStats_1.5.0          kableExtra_1.4.0
#> [17] BiocStyle_2.38.0           testthat_3.3.2
#>
#> loaded via a namespace (and not attached):
#> [1] Rdpack_2.6.6                sandwich_3.1-1              rlang_1.2.0
#> [4] magrittr_2.0.5              multcomp_1.4-30            otel_0.2.0
#> [7] compiler_4.5.3              mgcv_1.9-4                  systemfonts_1.3.2
#> [10] vctrs_0.7.3                 stringr_1.6.0              pkgconfig_2.0.3
#> [13] fastmap_1.2.0               magick_2.9.1               ellipsis_0.3.3
#> [16] backports_1.5.1             XVector_0.50.0             labeling_0.4.3
#> [19] rmarkdown_2.31              sessioninfo_1.2.3          tzdb_0.5.0
#> [22] nloptr_2.2.1                tinytex_0.59               purrr_1.2.2
#> [25] xfun_0.58                   cachem_1.1.0               jsonlite_2.0.0
#> [28] DelayedArray_0.36.1         BiocParallel_1.44.0        broom_1.0.13
#> [31] parallel_4.5.3              R6_2.6.1                   stringi_1.8.7
#> [34] bslib_0.11.0                RColorBrewer_1.1-3         car_3.1-5
#> [37] boot_1.3-32                 pkgload_1.5.2              brio_1.1.5
#> [40] jquerylib_0.1.4             estimability_1.5.1         Rcpp_1.1.1-1.1
#> [43] bookdown_0.46               usethis_3.2.1              zoo_1.8-15
#> [46] readr_2.2.0                 Matrix_1.7-5               splines_4.5.3
#> [49] tidyselect_1.2.1            rstudioapi_0.18.0          abind_1.4-8
#> [52] yaml_2.3.12                 codetools_0.2-20           pkgbuild_1.4.8
#> [55] lattice_0.22-9              tibble_3.3.1               plyr_1.8.9
#> [58] withr_3.0.3                 S7_0.2.2                   evaluate_1.0.5
#> [61] desc_1.4.3                  survival_3.8-6             xml2_1.5.2
#> [64] pillar_1.11.1               BiocManager_1.30.27        carData_3.0-6
#> [67] reformulas_0.4.4            rprojroot_2.1.1            hms_1.1.4
#> [70] scales_1.4.0                minqa_1.2.8                ARTool_0.11.2
#> [73] xtable_1.8-8                glue_1.8.1                 pheatmap_1.0.13
#> [76] emmeans_2.0.3               tools_4.5.3                lme4_2.0-1
#> [79] fs_2.1.0                    mvtnorm_1.3-6              cowplot_1.2.0
#> [82] grid_4.5.3                  tidyr_1.3.2                rbibutils_2.4.1
#> [85] devtools_2.5.2              nlme_3.1-169               Formula_1.2-5

```

```
#> [88] cli_3.6.6          textshaping_1.0.5   viridisLite_0.4.3
#> [91] S4Arrays_1.10.1    suglite_2.2.2       geepack_1.3.13
#> [94] gtable_0.3.6       sass_0.4.10         digest_0.6.39
#> [97] SparseArray_1.10.10 TH.data_1.1-5       farver_2.1.2
#> [100] memoise_2.0.1      htmltools_0.5.9     lifecycle_1.0.5
#> [103] MASS_7.3-65
```